CHAPTER 3

Building a Performance Testing Database

In the previous chapter you started working directly with PostgreSQL from psql and learned about the Rideshare application. You ran meta commands, inspected system catalogs, and configured an extension.

Moving along, you're about to learn to develop your own copy of your production database where you can begin to experiment with more extensive types of changes including structure modifications. You can also use this database for Performance testing, so it will be called the Performance database.

You'll begin to use advanced PostgreSQL concepts like *Functions* and *Procedures* by solving the very common problem faced by development teams, which is how they can get production-like data in their pre-production environment, without compromising any sensitive customer data. Some of the reasons to have production-like data are for performance testing of queries or for experimenting with indexes or different data model designs.

You'll write custom functions using SQL and learn about the *PL/pgSQL* Procedural Language. If you've never worked with this before don't worry, no prior knowledge is expected.

As your application scales up to higher amounts of queries and tables with greater row counts, having a place to test query changes and structure modifications becomes even more valuable.

To get started you'll download a pre-made database dump file for Rideshare. The database dump includes the structure and database objects but it also includes simulated sensitive customer information. Rideshare is fictional and the generated users are fictional as well, but you can treat these as if they are real users like you might find in a production database dump from your application.